

Dalton Kutzen

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Education

Brigham Young University — B.S. Molecular Biology
Provo, Utah

Expected Apr 2028

Experience

Bioinformatics Researcher

Aug 2026 – Present

Terooatea Lab, Brigham Young University · Provo, UT

- Developed computational methods and benchmark evaluations for high-dimensional single-cell RNA-seq (scRNA-seq) and pooled Perturb-seq screens (anchor-op).
- Implemented target-grouped nested cross-validation and matched signal-to-noise ratio (SNR) controls in Python to evaluate prediction performance against baseline models.
- Co-authored bioRxiv research on perturbation screen modeling and computational reproducibility.

Research Assistant

Sep 2025 – Present

Grose Lab, Brigham Young University · Provo, UT

- Prepared media, cultured bacteriophages with bacterial hosts, and conducted plaque and spot assays to evaluate phage activity and host interactions.
- Performed bioinformatic analyses to annotate and compare phage genomes, creating the **Phantom** Snakemake pipeline for reproducible annotation.
- Integrated wet lab microbiology techniques with computational genomics to investigate phage diversity, host range, and therapeutic applications.

IT Service Desk Supervisor

May 2026 – Present

College of Life Sciences, Brigham Young University · Provo, UT

- Mentor student technicians in troubleshooting best practices and train new hires on Life Sciences IT infrastructure, tools, and clear user communication.
- Drove internal automation, documentation, and reporting initiatives that reduced repetitive workflows and standardized support delivery college-wide.

IT Service Desk Technician

Aug 2025 – May 2026

- Oversaw deployment, security, and maintenance of 1,000+ campus workstations and lab computers, Supporting a community of 5,000+ users

Selected Projects

Phantom — Bacteriophage Analysis Workflow

Jun 2026 – Present

- Modular Snakemake pipeline for phage genome annotation and downstream comparative analysis.
- Uses isolated Conda environments and declarative workflow rules to make runs reproducible across systems.

Skills

Area	Tools & Proficiencies
Computational Biology	Multi-omics analysis, comparative genomics, Snakemake, Conda, Biopython
Wet Lab & Molecular Biology	Phage and bacterial culturing, plaque & spot assays, host range testing, media preparation, agarose gel electrophoresis
Programming & Data	Python, R, Bash, SQL, Git, pandas, tidyverse, ggplot2, Quarto
IT & Systems Infrastructure	IT Service Management (ITSM), Linux server administration, Proxmox, Docker, enterprise end-point management, networking, project management, technical mentoring, documentation